$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\left.\frac{\partial u}{\partial x}\right|_0 = \left.\frac{\partial u}{\partial x}\right|_a = 0 \quad , \quad u(x,0) = x$$

$$u(x,b) = 0$$

$u = \phi(x)\,\psi(y)$

$$\Rightarrow \quad \psi(y)\frac{\partial^2 \phi}{\partial x^2} + \phi(x)\frac{\partial^2 \psi}{\partial y^2} = 0$$

$$\Rightarrow \quad \frac{1}{\phi}\frac{\partial^2 \phi}{\partial x^2} = -\frac{1}{\psi}\frac{\partial^2 \psi}{\partial y^2} = -\lambda$$

Solving x-direction

$$\Rightarrow \quad \frac{\partial^2 \phi}{\partial x^2} + \lambda\,\phi(x) = 0$$

$$\left.\frac{\partial \phi}{\partial x}\right|_0 = \left.\frac{\partial \phi}{\partial x}\right|_a = 0$$

$\lambda = 0 \Rightarrow \phi(x) = C_1 x + C_2$

Applying B.C $\Rightarrow \quad \phi(x) = C_2$

$\lambda > 0, \quad \phi(x) = C_1 \cos(\sqrt{\lambda}\,x) + C_2 \sin(\sqrt{\lambda}\,x)$

$$\phi' = -\sqrt{\lambda}\, c_1 \sin(\sqrt{\lambda}\, x) + c_2 \sqrt{\lambda} \cos(\sqrt{\lambda}\, x)$$

$$\phi'|_0 = c_2 \sqrt{\lambda} = 0 \Rightarrow c_2 = 0$$

$$\phi'|_a = -\sqrt{\lambda}\, c_1 \sin(\sqrt{\lambda}\, a) = 0$$

$$\Rightarrow \sqrt{\lambda}\, a = n\pi \Rightarrow \lambda = n^2\pi^2/a^2$$

$\lambda < 0$ $\quad \phi(x) = c_1 \sinh(\sqrt{\lambda}\, x) + c_2 \cosh(\sqrt{\lambda}\, x)$

$$\phi' = c_1 \sqrt{\lambda} \cosh(\sqrt{\lambda}\, x) + c_2 \sqrt{\lambda} \sinh(\sqrt{\lambda}\, x)$$

$$\phi'|_0 = c_1 \sqrt{\lambda} = 0 \Rightarrow c_1 = 0$$

$$\phi'|_a = c_2 \sqrt{\lambda} \sinh(\sqrt{\lambda}\, x) = 0$$

$\Rightarrow c_2 = 0$ since λ must be zero to satisfy this equation with non trivial c_2 which is a contradiction.

Now for the y-direction we find solutions for corresponding eigenvalues we found in x.

Note we only have on homogenous BC

$$\Psi(b) = 0$$

$\lambda = 0$,

$$\psi(y) = c_1 y + c_2$$

$$\psi(b) = c_1 b + c_2 = 0$$

$$\Rightarrow \quad c_2 = -c_1 b$$

$$\Rightarrow \quad \psi(y) = c_1 y + c_2 = c_1 y - c_1 b$$

$$= c_1 (y - b)$$

$\lambda = 0$

for $\boxed{\lambda = n^2\pi^2/a^2}$:

$$\Rightarrow \quad \psi(y) = c_2 \cosh\left(\frac{n\pi}{a} y\right) + \boxed{c_3} \sinh\left(\frac{n\pi}{a} y\right)$$

$$\psi(b) = c_2 \cosh\left(\frac{n\pi}{a} b\right) + c_3 \sinh\left(\frac{n\pi}{a} b\right)$$

$$= 0$$

$$\Rightarrow \quad \boxed{c_3} = -c_2 \frac{\cosh(n\pi/a\, b)}{\sinh(n\pi/a\, b)}$$

$$\Rightarrow \psi(y) = c_2 \left[\cosh\left(\frac{n\pi}{a} y\right) - \underbrace{\frac{\cosh\left(\frac{n\pi}{a} b\right)}{\sinh\left(\frac{n\pi}{a} b\right)}}_{\text{constant}} \sinh\left(\frac{n\pi}{a} y\right)\right]$$

$$\Rightarrow u(x,y) = \sum_{\lambda} \psi_{\lambda} \phi_{\lambda}$$

$\lambda = 0$

$$= A_0 (y-b) + \sum_{n=1}^{\infty} A_n \overbrace{\cos\left(\frac{n\pi}{a}x\right)}^{\phi(x)} \times \underbrace{\left[\cosh\left(\frac{n\pi}{a}y\right) - \frac{\cosh\left(\frac{n\pi}{a}b\right)}{\sinh\left(\frac{n\pi}{a}b\right)} \sinh\left(\frac{n\pi}{a}y\right)\right]}_{\psi(y)}$$

lastly use that non homgenus B.C

$$u(x,0) = x$$

To find the coefficients